

Kalman filter can be found in many references, including [Rho71], [Spa88, Chapter 6], [Ho64, Wes85], [Kit96a, Chapter 6]. When the Kalman filter is derived this way, then no conclusions can be drawn about the optimality of the filter when the noise is not Gaussian. In fact, other optimal (nonKalman) filters have been derived for other noise distributions [Ser81]. Nevertheless, the Bayesian derivation proves that when the noise is Gaussian, the Kalman filter is the optimal filter. However, the least squares derivation that we used in Chapter 5 shows that the Kalman filter is the optimal *linear* filter, regardless of the pdf of the noise.

## 15.2 PARTICLE FILTERING

In this section, we derive the basic idea of the particle filter. The particle filter was invented to numerically implement the Bayesian estimator of the previous section. The main idea is intuitive and straightforward. At the beginning of the estimation problem, we randomly generate a given number  $N$  state vectors based on the initial pdf  $p(x_0)$  (which is assumed to be known). These state vectors are called particles and are denoted as  $x_{0,i}^+$  ( $i = 1, \dots, N$ ). At each time step  $k = 1, 2, \dots$ , we propagate the particles to the next time step using the process equation  $f(\cdot)$ :

$$x_{k,i}^- = f_{k-1}(x_{k-1,i}^+, w_{k-1}^i) \quad (i = 1, \dots, N) \quad (15.17)$$

where each  $w_{k-1}^i$  noise vector is randomly generated on the basis of the known pdf of  $w_{k-1}$ . After we receive the measurement at time  $k$ , we compute the conditional relative likelihood of each particle  $x_{k,i}^-$ . That is, we evaluate the pdf  $p(y_k | x_{k,i}^-)$ . As discussed in Section 15.1, this can be done if we know the nonlinear measurement equation and the pdf of the measurement noise. For example, if an  $m$ -dimensional measurement equation is given as  $y_k = h(x_k) + v_k$  and  $v_k \sim N(0, R)$  then a relative likelihood  $q_i$  that the measurement is equal to a specific measurement  $y^*$ , given the premise that  $x_k$  is equal to the particle  $x_{k,i}^-$ , can be computed as follows [compare with Equation (2.73)].

$$\begin{aligned} q_i &= P[(y_k = y^*) | (x_k = x_{k,i}^-)] \\ &= P[v_k = y^* - h(x_{k,i}^-)] \\ &\sim \frac{1}{(2\pi)^{m/2} |R|^{1/2}} \exp\left(\frac{-[y^* - h(x_{k,i}^-)]^T R^{-1} [y^* - h(x_{k,i}^-)]}{2}\right) \end{aligned} \quad (15.18)$$

The  $\sim$  symbol in the above equation means that the probability is not really given by the expression on the right side, but the probability is directly proportional to the right side. So if this equation is used for all the particles  $x_{k,i}^-$  ( $i = 1, \dots, N$ ), then the *relative* likelihoods that the state is equal to each particle will be correct. Now we normalize the relative likelihoods obtained in Equation (15.18) as follows.

$$q_i = \frac{q_i}{\sum_{j=1}^N q_j} \quad (15.19)$$

This ensures that the sum of all the likelihoods is equal to one. Next we resample the particles from the computed likelihoods. That is, we compute a brand new set of particles  $x_{k,i}^+$  that are randomly generated on the basis of the relative likelihoods  $q_i$ .

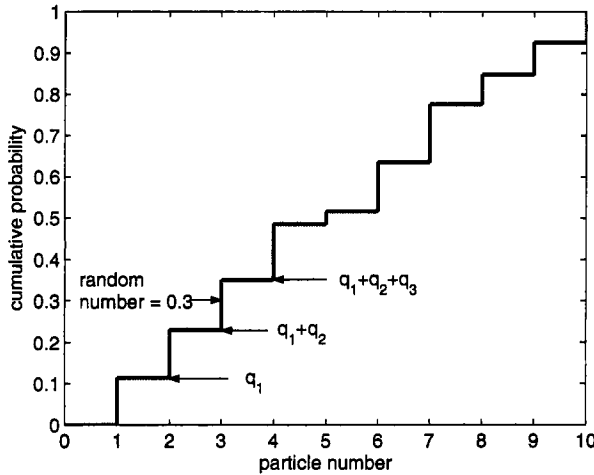
This can be done several different ways. One straightforward (but not necessarily efficient) way is the following [Ris04]. For  $i = 1, \dots, N$ , perform the following two steps.

1. Generate a random number  $r$  that is uniformly distributed on  $[0, 1]$ .
2. Accumulate the likelihoods  $q_i$  into a sum, one at a time, until the accumulated sum is greater than  $r$ . That is,  $\sum_{m=1}^{j-1} q_m < r$  but  $\sum_{m=1}^j q_m \geq r$ . The new particle  $x_{k,i}^+$  is then set equal to the old particle  $x_{k,j}^-$ .

This resampling idea is formally justified in [Smi92], where it is shown that the ensemble pdf of the new particles  $x_{k,i}^+$  tends to the pdf  $p(x_k|y_k)$  as the number of samples  $N$  approaches  $\infty$ . The resampling step can be summarized as follows:

$$x_{k,i}^+ = x_{k,j}^- \text{ with probability } q_j \quad (i, j = 1, \dots, N) \quad (15.20)$$

This is illustrated in Figure 15.2.



**Figure 15.2** Illustration of resampling in the particle filter. For example, if a random number  $r = 0.3$  is generated (from a distribution that is uniform on  $[0, 1]$ ), the smallest value of  $j$  for which  $\sum_{m=1}^j q_m \geq r$  is  $j = 3$ . Therefore the resampled particle is set equal to  $x_{k,3}^-$ .

The computational effort of the particle filter is often a bottleneck to its implementation. With this in mind, more efficient resampling methods can be implemented, such as order statistics [Car99, Rip87], stratified sampling and residual sampling [Liu98], and systematic resampling [Kit96]. Other ways of resampling have also been proposed [Mul91]. For example, the *a priori* samples  $x_{k,j}^-$  ( $j = 1, \dots, N$ ) could be accepted as *a posteriori* samples with a probability that is proportional to  $q_j$ . However, in this case additional logic must be incorporated to maintain a constant sample size  $N$ .

Now we have a set of particles  $x_{k,i}^+$  that are distributed according to the pdf  $p(x_k|y_k)$ . We can compute any desired statistical measure of this pdf. For example,

if we want to compute the expected value  $E(x_k|y_k)$  then we can approximate it as the algebraic mean of the particles:

$$E(x_k|y_k) \approx \frac{1}{N} \sum_{i=1}^N x_{k,i}^+ \quad (15.21)$$

The particle filter can be summarized as follows.

### The particle filter

1. The system and measurement equations are given as follows:

$$\begin{aligned} x_{k+1} &= f_k(x_k, w_k) \\ y_k &= h_k(x_k, v_k) \end{aligned} \quad (15.22)$$

where  $\{w_k\}$  and  $\{v_k\}$  are independent white noise processes with known pdf's.

2. Assuming that the pdf of the initial state  $p(x_0)$  is known, randomly generate  $N$  initial particles on the basis of the pdf  $p(x_0)$ . These particles are denoted  $x_{0,i}^+$  ( $i = 1, \dots, N$ ). The parameter  $N$  is chosen by the user as a trade-off between computational effort and estimation accuracy.
3. For  $k = 1, 2, \dots$ , do the following.

- (a) Perform the time propagation step to obtain *a priori* particles  $x_{k,i}^-$  using the known process equation and the known pdf of the process noise:

$$x_{k,i}^- = f_{k-1}(x_{k-1,i}^+, w_{k-1}^i) \quad (i = 1, \dots, N) \quad (15.23)$$

where each  $w_{k-1}^i$  noise vector is randomly generated on the basis of the known pdf of  $w_{k-1}$ .

- (b) Compute the relative likelihood  $q_i$  of each particle  $x_{k,i}^-$  conditioned on the measurement  $y_k$ . This is done by evaluating the pdf  $p(y_k|x_{k,i}^-)$  on the basis of the nonlinear measurement equation and the pdf of the measurement noise.
- (c) Scale the relative likelihoods obtained in the previous step as follows:

$$q_i = \frac{q_i}{\sum_{j=1}^N q_j} \quad (15.24)$$

Now the sum of all the likelihoods is equal to one.

- (d) Generate a set of *a posteriori* particles  $x_{k,i}^+$  on the basis of the relative likelihoods  $q_i$ . This is called the resampling step (for example, see Figure 15.2).
- (e) Now that we have a set of particles  $x_{k,i}^+$  that are distributed according to the pdf  $p(x_k|y_k)$ , we can compute any desired statistical measure of this pdf. We typically are most interested in computing the mean and the covariance.